

Theorem: Let (x, d) be a metric space such that every Cauchy seq in x has a convergent subsequence. Then (x, d) is complete.

Pf: Let $\{x_n\}_n$ be a Cauchy sequence, and let $\{x_{n_i}\}$ be a subsequence of $\{x_n\}_n$ that converges to x .

Claim: $x_n \rightarrow x$. Let $\varepsilon > 0$, $\exists N$ s.t. if $m, n > N$, then

$d(x_m, x_n) < \frac{\varepsilon}{2}$. Since $x_{n_i} \rightarrow x$, there is a natural

number i s.t. $i > N$ & $d(x_{i_p}, x) < \frac{\varepsilon}{2}$.

Then $d(x_n, x) \leq d(x_n, x_{i_p}) + d(x_{i_p}, x) < \varepsilon \quad \forall n > N$.

□

- If d is the usual metric on $\mathbb{R}$, then $(\mathbb{R}, d)$ is complete.

§ Continuous function:

Recall:

Thm: Let (x, τ) & $(Y, \mathcal{U})$ be topological spaces & let $f: x \rightarrow Y$. Then the following statements are equivalent:

(a) f is continuous

(b) If $V \in \mathcal{U}$, then $f^{-1}(V) \in \tau$.

(c) If C is a closed subset of $(Y, \mathcal{U})$, then $f^{-1}(C)$ is closed subset of (x, τ) .

Pf:

(b) $\Rightarrow$ (c). Let C be a closed subset of $(Y, \mathcal{U})$.

Then $Y - C \in \mathcal{U}$, by (b) $f^{-1}(Y - C) \in \tau$.

now, $f^{-1}(C) = X - f^{-1}(Y - C)$. Therefore $f^{-1}(C)$

is a closed subset of (x, τ) .

§ Compactness

Bolzano-Weierstrass theorem:

Every infinite ^{bounded} sequence in $\mathbb{R}^n$ has a convergent subsequence. Equivalently, a subset of $\mathbb{R}^n$ is sequentially compact iff it is closed and bounded.

Heine-Borel Theorem: Let a and b are real numbers with $a < b$ and $\mathcal{Q}$ is a collection of open intervals such that $[a, b] \subseteq \bigcup_{O \in \mathcal{Q}} O$, then there is a finite subset $\{O_1, O_2, \dots, O_N\}$ of $\mathcal{Q}$ such that

$$[a, b] \subset \bigcup_{n=1}^N O_n \quad \text{"N"-finite}$$

Defn: A collection $\mathcal{A}$ of subsets of a topological space (X, τ) is a cover of a subset B of X provided $B \subseteq \bigcup_{A \in \mathcal{A}} A$ and the cover $\mathcal{A}$ of B is an open cover of B provided each member of $\mathcal{A}$ is open ($\in \tau$).

Defn: (Compactness): A subset $\underline{\underline{A}}$ of a topological space (X, τ) is compact provided every open cover of A has a finite subcover.

Examples:

1. If X is a finite set & τ is any topology on X , then (X, τ) is compact.
2. $A = [a, b]$. Let $\mathcal{U}$ be an open cover of a closed interval $[a, b]$. For each $x \in [a, b]$, $U_x \in \mathcal{U}$ s.t. $x \in U_x$. Then there is an open interval I_x s.t. $x \in I_x \subseteq U_x$. By the Heine-Borel Theorem, there is a finite sub-collection $\{I_{x_1}, \dots, I_{x_n}\}$ s.t. $[a, b] \subseteq \bigcup I_{x_i}$. The collection $\{U_{x_1}, U_{x_2}, \dots, U_{x_n}\}$ of members of $\mathcal{U}$ that corresponds to $I_{x_1}, I_{x_2}, \dots, I_{x_n}$ is a finite sub-collection of $\mathcal{U}$ that cover $[a, b]$. Therefore $[a, b]$ is cpt.
3. The open interval $(0, 1)$ is not cpt because $\{(\frac{1}{n}, 1) : n \in \mathbb{N}\}$ is an open cover that does not have a finite sub-cover.
4. $\mathbb{R}$ is not cpt because $\{(-n, n) : n \in \mathbb{N}\}$ is an open cover that does not have a finite sub-cover.

Defn Let (X, d) be a metric space & let $\varepsilon > 0$. A finite subset F of X is called an ε -net for X if for each $x \in X$, there exists $p \in F$ s.t. $d(x, p) < \varepsilon$.
A metric space (X, d) is totally bounded if for each $\varepsilon > 0$, there is an ε -net for X .

Thm: Every totally bounded metric space is bounded.

Pf: Let (X, d) be totally bounded metric space.

Let F be $\frac{1}{2}$ -net for X & let $M = \max\{d(p, q) : p, q \in F\}$.

Let $x, y \in X$. Then there exists $a, b \in F$ such that $d(x, a) < \frac{1}{2}$, & $d(y, b) < \frac{1}{2}$.

Therefore, $d(x, y) \leq d(x, a) + d(a, b) + d(b, y) < M + 2$.

Hence, $d(x, y) < M + 2$ for all $x, y \in X$ Hence.

(X, d) is bounded.

Example:

For $x, y \in \mathbb{R}$, let $d(x, y) = \min\{|x - y|, 1\}$. Then $(\mathbb{R}, d)$ is bounded metric space, but it is not totally bounded.

Since $d(x, y) \leq 1 \forall x, y \in \mathbb{R} \Rightarrow (\mathbb{R}, d)$ is bounded.

$(\mathbb{R}, d)$ is not totally bounded, as there is no $\frac{1}{2}$ -net

for $\mathbb{R}$. Let F be any finite subset of $\mathbb{R}$ & let

p be the largest member of F . If $q \in F$, then

$d(p+1, q) = 1$ & hence there is not $\frac{1}{2}$ -net for $\mathbb{R}$.

Theorem: A metric space (X, d) is cpt iff it is complete and totally bounded.

Defn: A topological space (X, τ) is sequentially cpt provided every seq in X has a subseq that converges.

Defn. Let (X, d) & (Y, ρ) be metric spaces & let $f: X \rightarrow Y$ be a function. Then f is uniformly continuous if for each $\varepsilon > 0$, there is a positive number δ such that if $x, y \in X$ & $d(x, y) < \delta$

$$\Rightarrow \rho(f(x), f(y)) < \varepsilon.$$

Thm. Let (X, d) be a cpt metric space, let (Y, ρ) be a metric space, and let $f: X \rightarrow Y$ be a continuous fn. Then f is uniformly continuous.

Thm. The continuous image of a compact space is compact.

pf. Let (X, τ) be a compact space, let (Y, τ') be a top. space, let $f: X \rightarrow Y$ be a continuous fn. that maps X onto Y , & let $\mathcal{O}$ be an open cover of Y . Then $\mathcal{O}' = \{f^{-1}(U) : U \in \mathcal{O}\}$ is an open cover of X . Since X is cpt, $\mathcal{O}'$ has a finite sub-cover $\mathcal{O}''$. Thus $\{U \in \mathcal{O} : f^{-1}(U) \in \mathcal{O}''\}$ is a finite subcollection of $\mathcal{O}$ that covers Y . Therefore Y is cpt.

$f: [a, b] \rightarrow \mathbb{R}$ is continuous $\Rightarrow f$ is uniformly continuous.
 $\underbrace{\hspace{1cm}}_{\text{cpt set in } \mathbb{R}}.$

For each $x \in [a, b]$ & given $\varepsilon > 0$, there exists $\delta_x > 0$ s.t.

$$|x - y| < \delta_x \Rightarrow |f(x) - f(y)| < \frac{\varepsilon}{2}.$$

The family $\left\{ \left(x - \frac{\delta_x}{2}, x + \frac{\delta_x}{2} \right) \right\}_{x \in [a, b]}$

is an open cover of $[a, b]$

By compactness of $[a, b]$, we can extract a finite-subcover:

$$[a, b] \subseteq \bigcup_{i=1}^N \left(x_i - \frac{\delta_{x_i}}{2}, x_i + \frac{\delta_{x_i}}{2} \right)$$

Choose $\delta = \min_{1 \leq i \leq N} \frac{\delta_{x_i}}{2} > 0$

Now, if $|x - y| < \delta$, then x, y lie in one of the intervals of the finite sub-cover, a ray around x_i .

Thus by Local continuity of x_i

$$\Rightarrow |f(x) - f(y)| < \varepsilon.$$

Hence f is uni-f. continuous.

Connectedness

From an intuitive point of view we say that something is connected if it consists of one piece. Spaces that are not connected are also interesting.

Connectedness was defined for bounded, closed subsets of $\mathbb{R}^n$ by Cantor in 1883, but his definition is not suitable for general topological spaces. In 1892, Camille Jordan gave a different definition of Connectedness for bounded, closed subsets of $\mathbb{R}^n$.

Defn: A topological space is connected if it cannot be expressed as the union of two non-empty, disjoint open sets.

Defn: Let (X, τ) be a topological space and let A and B be subsets of X . Then A and B are separated in X if $\bar{A} \cap B = A \cap \bar{B} = \emptyset$. If A and B are separated in X and $X = A \cup B$, then A and B are said to form a separation of X .

Theorem: A topological space (X, τ) is connected iff it cannot be expressed as the union of two non-empty sets that are separated in X .

pf: Suppose X is not connected. Then there are non-empty, disjoint open sets U & V such that $X = U \cup V$. Then U and V are also closed,

so $\bar{U} \cap V = U \cap \bar{V} = \emptyset$ and $U \cap \bar{V} = U \cap V = \emptyset$.

Therefore U & V are separated in X .

Suppose now that there are non-empty subsets A and B such that $X = A \cup B$ & $\bar{A} \cap B = A \cap \bar{B} = \emptyset$

Since $X = A \cup B$ & $\bar{A} \cap B = \emptyset \Rightarrow \bar{A} \subseteq A$, so

A is closed. The same argument shows that B is closed. Therefore A and B are also open, & hence X is not connected.

Corr. If A & B form a separation of the topological space (X, τ) , then A and B are both open and closed.

Example:

1. Let X be a set with more than one member, & let τ be the discrete topology on X . Then (X, τ) is not connected.
2. Let $X = \mathbb{R} \setminus \{0\}$ & let τ be the usual topology on X . Since $X = (-\infty, 0) \cup (0, \infty)$, (X, τ) is not connected.
3. $\mathbb{A}^* = \mathbb{Q}$, $X = (\mathbb{R}, \tau)$ usual topology. Connected or disconnected?
4. Let τ be the usual topology on $\mathbb{R}$. Then $(\mathbb{R}, \tau)$ is connected.

pf. Suppose $(\mathbb{R}, \tau)$ is not connected. Then there exist disjoint non-empty open sets U & V such that $\mathbb{R} = U \cup V$. Since $U = \mathbb{R} \setminus V \subseteq V = \mathbb{R} \setminus U$, U & V are also closed.

Let $a \in U$ and let $b \in V$. W.L.O.G, we may assume that $a < b$. Let $W = U \cap [a, b]$. Since W is bounded it has least upper bound c . Since W is closed, $c \in W$. Since $W \cap V = \emptyset$, $c \neq b$. Since c is the L.U.B of W , $(c, b] \subseteq V$. Therefore $c \in \overline{V}$.

But V is closed, so $c \in V$. Therefore $c \in U \cap V \rightarrow \leftarrow$

5. Let τ be the usual topo. Then a subset $(\mathbb{R}, \tau)$ is connected iff it is an interval.

Qn: Continuous Image of $\mathbb{R}$?

Theorem: Let (X, τ) be a connected space, let $(Y, \mathcal{U})$ be a topological space, and let $f: X \rightarrow Y$ be a continuous function that maps X onto Y . Then $(Y, \mathcal{U})$ is connected.

Pf: Suppose $(Y, \mathcal{U})$ is not connected. Then there are disjoint non-empty open sets U & V such that $Y = U \cup V$. Since f is continuous $\Rightarrow f^{-1}(U)$ & $f^{-1}(V)$ are non-empty open subsets of X .

Since $f^{-1}(U) \cap f^{-1}(V) = \emptyset$ & $X = f^{-1}(U) \cup f^{-1}(V)$
 $\Rightarrow (X, \tau)$ is not connected. $\rightarrow \leftarrow$

Corr: Let $f: X \rightarrow \mathbb{R}$ be a continuous function, let $a, b \in X$ s.t. $f(a) < f(b)$, & let $d \in \mathbb{R}$ s.t. $f(a) < d < f(b)$. Then there exists $c \in X$ s.t. $f(c) = d$.

pf. $f(x)$ is a connected subset of $\mathbb{R}$. Moreover $f(x)$ is an interval. Therefore $[f(a), f(b)] \subseteq f(x)$. and so there exists $c \in X$ s.t. $f(c) = \theta$.

Example: $A = \left\{ (x, y) \in \mathbb{R}^2 : x = 0 \text{ \& } -1 \leq y \leq 1 \right\}$

$B = \left\{ (x, y) \in \mathbb{R}^2 : 0 < x \leq 1 \text{ \& } y = \sin\left(\frac{\pi}{x}\right) \right\}$

& let $X = A \cup B$. Let τ be the subspace topo. induced by the usual top on $\mathbb{R}^2$.

Then (X, τ) is connected.

pf. Define $f: (0, 1] \rightarrow B$ by $f(x) = \left(x, \sin\left(\frac{\pi}{x}\right)\right)$.

Then f is continuous maps $(0, 1]$ onto B . $\Rightarrow B$ is connected.

$\Rightarrow \bar{B}$ in $\mathbb{R}^2$ is connected. But $\bar{B} = X \Rightarrow X$ is also connected.

This is called topologist's sine curve.

§ Pathwise and Local Connectedness

Defn: A path in a topological space (X, τ) is a continuous function $f: I \rightarrow X$. The path " f " is called a path from $f(0)$ to $f(1)$.

Defn: A topological space (X, τ) is pathwise connected if for each pair of $a, b \in X$, there is a path f in X s.t. $f(0) = a$, $f(1) = b$.

Theorem: Every pathwise connected space is connected.

Pf.: Suppose there is a pathwise connected space (X, τ) that is not connected. Then $\exists U, V (\neq \emptyset)$

s.t. $X = U \cup V$. Let $a \in U$ & $b \in V$ & let f is a path from a to b . Then $f^{-1}(U)$ & $f^{-1}(V)$ are non-empty disjoint open subsets of I

s.t. $I = f^{-1}(U) \cup f^{-1}(V) \Rightarrow I$ is disconnected. $\rightarrow \leftarrow$.

Remark: topologist's sine curve is not pathwise connected.